

Manufacturing ERP Development Handbook

Portal map, page connections, process flows, team allocation and the 18-week plan to release 1.

SCREENS IN THE PRODUCT

253 across 15 portals

TEAM

**6 people (1 lead + 5
developers)**

REMAINING EFFORT

383 person-days

TARGET TO GO-LIVE

18 weeks

Version 1.0 · September 2026 · Internal — for the development team

Derived from the repository as it stands: `frontend/src/config/portals.ts`, `navigation.ts`, `App.tsx`, `backend/app` and `docs/srs`.

Contents

1	How to use this handbook	Scope, and the one assumption behind every number
2	The system in one page	What we are building, and the portal idea
3	Where the build stands today	Measured status of all 15 portals
4	Portal-wise pages and modules	Every menu group and screen, portal by portal
5	Page-to-page connections	Which screen opens which, inside each portal
6	Step-by-step process flows	The eight flows the business actually runs
7	Overall project workflow	Enquiry to cash, end to end
8	Dependencies between portals	What must exist before what
9	Development sequence	Six phases, and why in that order
10	Team allocation	Who works on which portal and pages
11	Responsibilities of each team member	Owned, shared and escalated
12	Duration estimates	Person-days per portal, and how they were built
13	The 18-week schedule	Week-by-week plan and milestones
14	Definition of done	The checklist every screen must pass
15	Risks and open items	What can break the plan, and the fix
A	Appendix A — Full screen and route reference	All 253 routes, by portal

1. How to use this handbook

This is the working document for the six of us building the SSB Manufacturing ERP. It answers five questions, in this order: what screens exist, how they connect, who builds what, in what order, and by when. Read sections 1 to 3 once. Then live in sections 10 to 14 — those are the ones you will come back to every week.

1.1 The one assumption behind every number

This plan estimates the work that remains, not the work from zero

The repository already contains a substantial, working product: 253 screens are routed, and 5 of the 15 portals are wired end to end to a real FastAPI backend. Every duration in section 12 is therefore **remaining effort to finish a portal**, not the effort to build it from scratch. Section 3 shows the measured status that this rests on, so you can check the basis yourself. If you want a from-zero estimate instead, say so and the numbers change substantially.

1.2 What is in scope for release 1

Thirteen portals. Two are deliberately deferred to release 2: **Sales & CRM** and the **Shop Floor Terminal**. Both are marked `status: 'PLANNED'` in `portals.ts`, neither has a single screen built, and Sales & CRM has no approved specification (SRS Volume 2 is "Not started"). Building them now would mean inventing business rules, which section 15 explains we must not do. The external Supplier and Customer portals and the React Native mobile app are also release 2.

1.3 How the estimates were produced

Every screen was classified into one of five types, each with a fresher-calibrated day rate that already assumes the shared component library in `frontend/src/components` is reused rather than rewritten. The per-portal totals in section 12 are the sum of their screens. The rates:

Type	What it is	Days per screen	Example
A	Lookup or simple master: list, form, soft delete, where-used	1.0	Reason codes, UOM
B	List and detail over an API that already exists	1.5	Category ledger
C	Dashboard or report over existing APIs	1.5	Portal dashboards
D	Transaction document: header, lines, state machine, approval, print	4.0	Material issue, maintenance work order
E	Calculation engine plus its screen	6.0	Payroll run, OEE, cost roll-up

A type D or E figure includes its backend: SQLAlchemy models, Alembic migration, repository, application service, Pydantic schemas, router, permissions, and tests. It does not include code review, which is the lead's time, or UAT, which is budgeted separately in section 12.

2. The system in one page

SSB Manufacturing ERP is one connected system for a stainless steel vacuum-insulated water bottle plant. It runs the business from buying steel coil, through production, quality and packing, to dispatch, invoicing and payroll. It is a **modular monolith**: a single deployable FastAPI application partitioned into modules that map one-to-one onto the SRS volumes, with one React front end on top.

2.1 What a portal is

A portal is the working context a person signs into — not a security boundary. A storekeeper receiving goods works in Inventory; a buyer works in Procurement. Permissions are resolved per user regardless of which portal they are in, server-side, every time. The portal only decides which part of the product they are put in front of, and which sidebar tree they get. This matters to you as a developer for one reason: **never treat the sidebar as a control**. Hiding a menu item is a convenience. The permission dependency on the endpoint is the control.

2.2 The stack, which is locked

Layer	Decision
Database	MySQL 8.0+, InnoDB, <code>utf8mb4_0900_ai_ci</code>
Backend	Python 3.12+, FastAPI, SQLAlchemy 2.0 typed <code>Mapped[]</code> , Alembic, Pydantic v2
Async and cache	Celery + Redis, <code>celery beat</code> ; Redis 7+ for cache, locks, rate limits
Front end	React 18, TypeScript, Vite, Tailwind, TanStack Query for server state, Zustand for client state
Forms and tables	React Hook Form + Zod; TanStack Table, virtualised
Reports	WeasyPrint for PDF, openpyxl / xlsxwriter for Excel
Auth	JWT RS256 access token, 15 minutes, plus rotating opaque DB-backed refresh token

Adding a runtime dependency, framework or datastore requires an ADR in [docs/adr/](#) and the lead's approval. This is not negotiable and it is not bureaucracy: a fresher adding a second state library or a second table component is the single fastest way to make 253 screens inconsistent.

2.3 The four rules you will break first

Rule	Why it exists
Soft delete only	Never <code>DELETE</code> a business row. Set <code>deleted_at</code> and <code>deleted_by</code> . Unique keys carry a generated <code>deleted_key</code> companion column because MySQL treats multiple NULLs as distinct.
Tenancy is application-enforced	MySQL has no row-level security. The base repository injects <code>company_id</code> into every query and every insert. Raw SQL must bind it explicitly; CI greps for <code>text(</code> without a tenant bind.
Public <code>uid</code> , never <code>id</code>	URLs and payloads use the ULID <code>uid</code> . The <code>BIGINT id</code> never leaves the database.
No <code>FLOAT</code> for money or quantity	Quantity and rate are <code>DECIMAL(18,6)</code> , amount is <code>DECIMAL(18,2)</code> , percentage is <code>DECIMAL(9,4)</code> . Round half-up once, at persistence.

3. Where the build stands today

This table was measured from the repository, not estimated. "Screens" counts distinct routes in `App.tsx` plus the 22 generic master routes driven by `masterRegistry.ts`. "Wired" counts page components that import from `@api`, meaning they talk to the real backend rather than a mock fixture.

253 SCREENS ROUTED	130 WIRED TO THE BACKEND	5 PORTALS COMPLETE	8 PORTALS STILL TO FINISH	
Portal	Screens	Wired	Status	Backend module and routers that exist
Administration & Platform	39	26	Part-built	iam, organisation, numbering, workflow, security, parameters, audit
Master Data	38	34	Complete	masters plus ~25 lookup routers
Procurement	8	8	Complete	procurement, rfq, quotation, purchase_order, grn, supplier
Inventory & Stores	31	16	Part-built	inventory: stock, txn, txn_doc, count, analysis, report
Product Engineering	13	12	Complete	8 engineering_* routers: BOM, routing, operations, work centres, tools, cost, changes, documents
Production Planning	11	4	Part-built	planning: MRP engine, capacity, demand, dashboard
Shop Floor Execution	17	0	UI only	— none —
Quality	11	11	Complete	inspections, ncr, capa, calibration, complaint, defects, quality_plans, supplier_quality
Plant Maintenance & EAM	12	0	UI only	— none —
Packing, Dispatch & Logistics	19	19	Complete	12 routers: packing order, carton, pallet, label, pick list, loading, shipment, POD, freight, exports, returns
Finance & Accounts	12	0	UI only	— none —
HR, Payroll & Workforce	22	0	UI only	— none —
Analytics & Insights	18	0	UI only	partial analytics, reporting
Sales & CRM	0	0	Release 2	— none, and no SRS —
Shop Floor Terminal	0	0	Release 2	— none —
Total	253	130	Plus sign-in and profile screens	

Read this before you feel behind

"UI only" is not zero. Those 81 screens already have their layout, their mock fixture and their TypeScript types written. The mock file in `frontend/src/mock/` is a working specification of the shape the API must return. Your job on those screens is to build the backend to match, then delete the mock. That is why five UI-only portals cost 220 person-days and not double that.

3.1 The framework services that are already done

Seven cross-cutting services carry every document in every portal, and all seven are built and working. You do not need to build them, you need to use them:

Service	Where it lives	What you must do
Authentication and RBAC	<code>modules/iam</code>	Declare a permission dependency on every endpoint. CI fails an endpoint with none.
Organisation and tenancy	<code>modules/organisation</code>	Resolve <code>TenantContext</code> ; let the base repository inject <code>company_id</code> .
Document numbering	<code>modules/numbering</code>	Ask the numbering engine for a document number. Never write a sequence in module code.
Workflow and approvals	<code>modules/workflow</code>	Create a workflow instance for every document, even when the rule is auto-approve.
Audit trail	<code>modules/security</code> , <code>audit</code>	Emit audit rows on create, update, submit, approve, reject, cancel, amend. Append-only.
Parameters	<code>modules/parameters</code>	Read configurable behaviour from parameters, not from constants in your code.
Reporting	<code>modules/reporting</code>	The one sanctioned place to join across module tables, read-only.

4. Portal-wise pages and modules

Each portal below lists its sidebar groups exactly as they are defined in `frontend/src/config/navigation.ts`, with the screens under each. The status column is per group: **Done** means wired to the backend, **Part** means some screens still read a mock fixture, **Shell** means the whole group is mock.

4.1 Administration & Platform Services — 39 screens Lead

Menu group	Screens	Status
Dashboard	Home dashboard /	Done
Organisation	Structure explorer, Company, Branches, Plants & lines, Warehouses & bins, Departments, Cost centres, Financial year & periods, Currency & rates (9)	Done
Access Control	Users, Roles, Permission explorer, Delegations, Sessions & devices, API keys, Segregation of duties, Login activity (8)	Done
Workflow	Approvals inbox, Approval matrix, Workflow designer, Workflow monitor (4)	Done
Platform Services	Document numbering done , Notifications, Master data framework, Attachments & DMS, Document repository, Import & export, Collaboration, Barcode & labels, Report engine (9)	Part
Compliance & Ops	Audit trail done , Security policy done , Logging & monitoring, Administration reports, System parameters done , Integrations, Background jobs, Backup & restore, Licence (9)	Part

Remaining work: 13 screens, all of them cross-cutting enablers that other portals consume — notifications, attachments, the import framework, the report engine, barcode and labels. This is why the lead owns this portal: finishing it unblocks everyone else.

4.2 Master Data — 38 screens D2

Menu group	Screens	Status
Business Partners	Customers, Suppliers, Transporters, Banks, Contact persons (5)	Done
Product Engineering	Item Category, Product Category, Product Items, Company Items, Products, Bottle models, Capacities, Colours & finishes, Lid & cap types, Packaging materials (10)	Done
Raw Material	Steel grades, Steel thickness (2)	Done
Production	Machines, Production lines, Work centres (<i>read-only, owned by Engineering</i>), Plants, Shifts, Holiday calendar, Planning policy (7)	Done
Quality	Inspection parameters, Defect types (2)	Done
HR	Employees (1)	Done
Finance & Tax	HSN / SAC codes, Tax rates, Payment terms, Currency & rates, Cost centres (5)	Done
Reference & Geography	Units of measure, Reason codes, Countries, States, Cities (5)	Done
Support screens	Masters overview, Duplicate finder, Import & export, Supplier forms (4)	Shell

The generic master pattern

22 of these screens are not hand-written. They are driven by a registry (`mock/masterRegistry.ts`) rendered through one `SimpleMasterPage` component. When you add a simple lookup master, add a registry entry — do not add a page. Only masters with real behaviour (Customer, Supplier, Item, Machine, Planning policy) get their own component.

4.3 Procurement — 8 screens Complete

A deliberately flat menu, in the exact order the work happens. Screens that support the chain but are not stages of it (rate contracts, imports, ASN, returns, supplier evaluation) stay reachable by URL and are opened from the screen that needs them.

Screen	Route	What it produces
Dashboard	/procurement	Spend, open PR/PO, overdue receipts
Purchase Requisition	/procurement/requisitions	PR document, from MRP or manual
Request for Quotation	/procurement/rfq	RFQ to a supplier set
Vendor Quotations	/procurement/quotations	Supplier quote capture
Quotation Comparison	/procurement/comparison	Landed-cost comparison and recommendation
Purchase Order	/procurement/orders	PO, amendable with revision history
Goods Receipt Note	/procurement/grn	GRN, which posts stock and triggers incoming QC
Approvals	/procurement/approvals	Procurement slice of the approvals inbox

4.4 Inventory & Stores — 31 screens D1

Area	Screens	Status
In the sidebar	Dashboard, Current Stock, Stock In, Stock Out, Stock Return, Stock Transfer, Batch & Expiry, Low Stock Monitor, Stock Ledger, Category Ledger, Warehouse Setup (11)	Part
Stock model	Stock, Ledger, Valuation, Ageing, ABC-XYZ, Reorder, Structure, Bin map, Warehouses (9)	Part
Movements	Receipts, Putaway, Issues, Returns, Transfers, Goods in transit, Adjustments, Scrap, Job work, Requisitions, Reservations, Movement (12)	Part
Identity & trace	Batches, Serials, Traceability, Labels (4)	Part
Counting	Counting, Count workspace, Verification, Variance (4)	Part
Reports	Reports (1)	Shell

Built: the stock balance model, the stock ledger, the transaction engine ([txn_router](#), [txn_doc_router](#)), cycle counting, ABC-XYZ analysis, stock reports, warehouse and bin setup. **Remaining, 21 screens:** issue, return, transfer, goods-in-transit, adjustment, scrap, put-away, job work, requisition, reservation, serials, traceability, labels, category ledger, verification, variance, dashboard and reports.

Why Inventory is the highest-leverage remaining portal

Nothing in Shop Floor Execution can be tested until material issue posts a real stock movement, and nothing in Finance can be reconciled until valuation posts to the ledger. Inventory is on the critical path for two other portals, which is why it starts in week 3 and is owned full-time.

4.5 Product Engineering — 13 screens Complete

Menu group	Screens
Dashboard	Engineering dashboard
Documents	Engineering documents, Document types
Bill of Materials	BOM register, BOM explorer (<i>multi-level tree</i>)
Routing	Manufacturing routing, Operations, Work centres, Tools & dies
Cost & Change	Cost roll-up, Engineering changes (ECN)
Reports	Engineering reports, Audit trail (<i>shared</i>)

Engineering owns the BOM, the routing and the work-centre record. The product master itself lives in Masters. Planning explodes the BOM; Shop Floor reads the routing. Both of those are consumers, so **the BOM and routing contracts must not change once MES starts in week 6.**

4.6 Production Planning — 11 screens D2

Screen	Route	Status	Note
Dashboard	/planning	Done	Plan health, exceptions
Sales Forecast	/planning/forecast	Shell	Needs seasonality factors
Demand	/planning/demand	Part	Backend exists, page is mock
Master Schedule	/planning/mps	Shell	Firms up net demand into a plan
MRP	/planning/mrp	Done	The engine. BOM explosion, buy/make split
Capacity Planning	/planning/capacity	Done	Work-centre load against calendar
Production Orders	/planning/orders	Shell	Release gate logic exists in tests
Schedule Board	/planning/schedule	Shell	Drag-and-drop sequencing
Material Reservation	/planning/reservations	Shell	Reserved-qty column already in DB
Production Calendar	/planning/calendar	Shell	Shift and holiday overlay
Reports	/planning/reports	Shell	

4.7 Shop Floor Execution — 17 screens D3 + D2

Menu group	Screens	Hardest part
Orders	Production orders, Work orders, Work centre queue	Order-to-work-order explosion from routing
Execution	Production entry, Work in progress, Work instructions	Production entry: barcode, backflush, yield and scrap in one transaction
Machines	Machine status, Downtime, OEE	OEE: availability × performance × quality, per shift and machine
Losses	Scrap, Rework	Scrap SS has recoverable value — it is an inventory item, not a write-off
People	Shift management, Labour tracking	Feeds HRMS incentives and labour cost
Traceability	Traveller card, Production genealogy	Heat number to coil to blank to body to bottle to carton
Reports	Production reports, Audit trail (<i>shared</i>)	

The whole portal is a shell today. It is also the largest single piece of remaining work, and it sits downstream of Planning production orders, Engineering routing and Inventory issues — which is exactly why it starts in week 6, not week 3.

4.8 Quality — 11 screens Complete

Flat menu: Dashboard, Inspection Plans, Inspections, Defects & Pareto, Non-Conformance (NCR), CAPA, Calibration, Supplier Quality, Customer Complaints, Audits, Reports. All eleven are wired. Quality is a gate in three separate flows — incoming on GRN, in-process on operation confirmation, final before packing — so treat its API as frozen.

4.9 Plant Maintenance & EAM — 12 screens D1

Flat menu: Dashboard, Asset Register, Preventive Plans, Work Orders, Breakdowns, Condition Monitoring, Spare Parts, Technicians, Safety Permits, Utilities, Shutdowns, Reports. Entirely a shell. Mostly master-plus-document work, with one genuinely hard screen: the preventive-plan scheduler, which must generate work orders on a calendar *and* a meter trigger.

4.10 Packing, Dispatch & Logistics — 19 screens Complete

Menu group	Screens
Packing	Packing orders, Packaging material, Cartons, Pallets, Labels
Dispatch	Dispatch planning, Pick lists, Vehicle loading, Vehicles
Logistics	Shipments, Shipment tracking, Proof of delivery, Returns, Freight & cost, Transporter performance
Export	Export shipments, Export documents
Reports	Dispatch reports, Audit trail (<i>shared</i>)

4.11 Finance & Accounts — 12 screens D4

Flat menu: Dashboard, Chart of Accounts, Journals & Ledger, Statements, Receivables, Payables, Banking, Cost & Variance, GST & TDS, Fixed Assets, Budgets, Period Close. Entirely a shell, and the highest-risk portal in the plan because correctness is not negotiable and two of its inputs (sales invoicing, payroll) are themselves unbuilt.

Indian statutory logic is isolated, always

GST, e-invoice IRN, e-way bill, TDS, PF and ESI logic goes behind a [statutory/](#) adapter, with rates and slabs held as configurable master data. Never hard-code a rate in business logic. Rates change by government notification; code must not.

4.12 HR, Payroll & Workforce — 22 screens D5 + D4

Menu group	Screens
People	Employees, Organisation structure
Recruitment	Manpower requisitions, Candidate pipeline
Time & Attendance	Attendance, Overtime, Shifts & roster, Leave
Payroll	Payroll run, Payslips, Salary structures, Statutory compliance
Workforce	Production incentives, Labour cost, Skill matrix, Contractor labour
Growth	Performance, Training
Self Service	My page, My team
Reports	HR reports, Audit trail (<i>shared</i>)

The biggest portal by screen count and the biggest by effort. Attendance import, the payroll engine and statutory returns are the three screens that decide whether this portal lands on time.

4.13 Analytics & Insights — 18 screens D3 + D4

Executive Overview, Dashboards, AI Insights, eight domain analytics pages (Production, Quality, Maintenance, Procurement, Inventory, Sales, Finance, Workforce), Forecasts, KPI library, Scorecards, Alerts & Exceptions, Report Catalogue, Data sources, Access & audit.

Read-only across every module, built in the [reporting](#) module with explicit SQL. This is the one sanctioned place in the codebase where a query may join across module tables. Two permissions gate the sensitive pages: [BI.FINANCIAL.VIEW](#) and [BI.PEOPLE.VIEW](#). It is scheduled last because a dashboard over an unbuilt module has nothing to show.

4.14 Deferred to release 2

Portal	Scope	Blocked by
Sales & CRM	Leads, enquiries, quotations, sales orders, invoices, complaints, customer ledger	SRS Volume 2 not written
Shop Floor Terminal	Scan-driven operator screens for line confirmation and material movement	Shop Floor Execution must exist first

Portal	Scope	Blocked by
Supplier Portal	RFQ response, PO acknowledgement, ASN, invoice status	External auth model undecided
Customer Portal	Order status, dispatch tracking, invoices, complaints	Depends on Sales & CRM

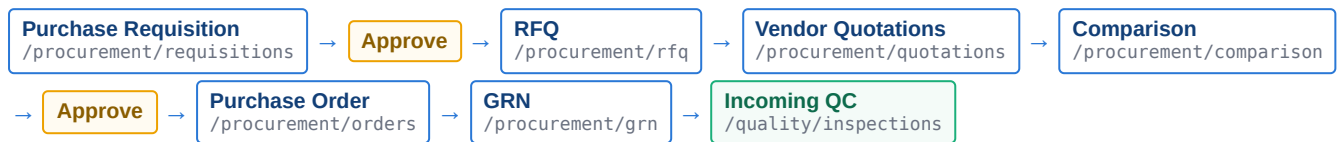
Sales & CRM leaves two holes you will hit in release 1

Planning normally nets demand against **sales orders**, and Finance receivables normally starts from a **sales invoice**. Neither exists. For release 1: Planning takes demand from manual entry and the forecast (the demand screen already supports this), and Finance raises the sales invoice from the dispatch shipment instead of from a sales order. Document both as temporary, in [docs/srs/open-questions.md](#).

5. Page-to-page connections

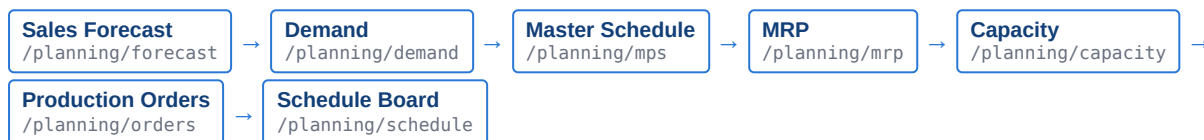
A screen is rarely a destination. Almost every screen in this product is reached *from* another screen, carrying a document reference with it. The chains below are the real ones — taken from the navigation comments and the cross-page navigation calls in the page components. Learn your portal's chain before you write a line of code, because the chain is what the user experiences.

5.1 Procurement

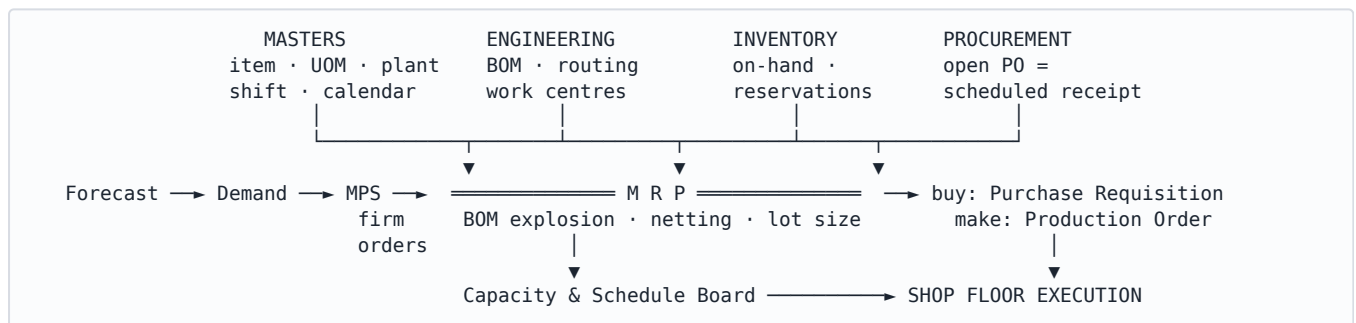


Side branches: a rejected GRN line opens **Purchase return**; an approved GRN posts to **Stock In**; MRP creates the PR without a human typing it.

5.2 Production Planning



MRP is the hub, and it reads from four other portals. Its outputs split two ways: a **make** line becomes a production order, a **buy** line becomes a purchase requisition in Procurement. Material reservation locks stock against the released order.



5.3 Inventory & Stores



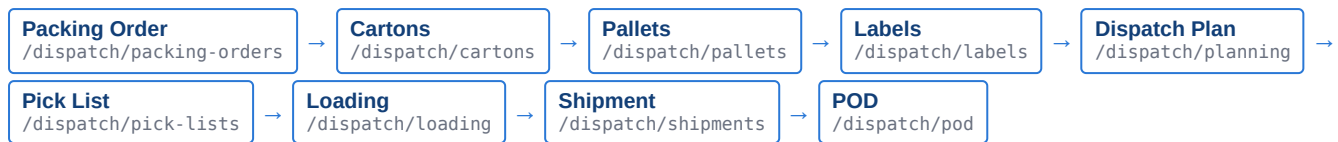
Every one of those movements writes one row to the **Stock Ledger** (`/inventory/ledger`), which is the audit spine of the portal: if a movement is not in the ledger, it did not happen. Batch and serial screens hang off the ledger and feed **Traceability**. Low Stock Monitor generates the requisitions that start section 5.1.

5.4 Shop Floor Execution



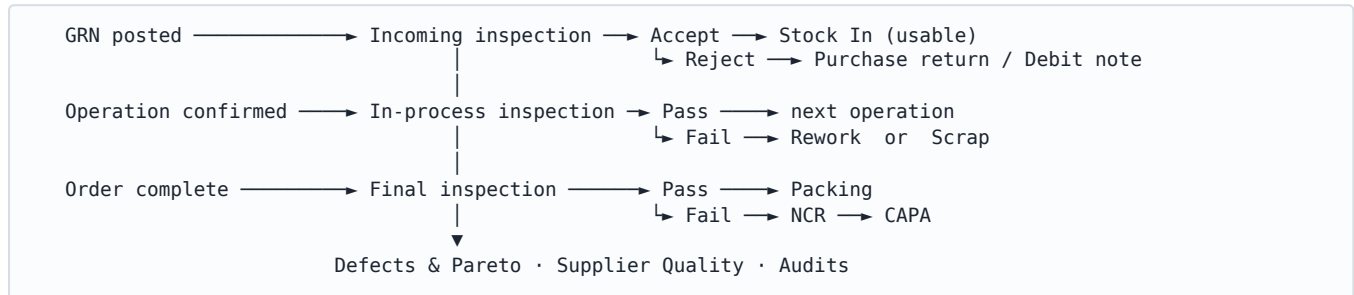
Production entry is the busiest screen in the product and the one with the most side effects. A single confirmation writes output, consumes material by backflush, records scrap and rework, stamps the traveller card, and updates OEE. Get its transaction boundary right and the rest of the portal is straightforward.

5.5 Packing, Dispatch & Logistics



Export orders branch at Shipment into **Export shipments** and **Export documents**. Freight cost and transporter performance are fed by the completed shipment, not entered separately.

5.6 Quality, which is a gate in three places



5.7 Finance and HR, which are mostly consumers

Screen	Is fed by
Payables /finance/payables	Three-way match of PO, GRN and purchase invoice
Receivables /finance/receivables	Sales invoice raised from the dispatch shipment
Cost & Variance /finance/costing	Engineering cost roll-up as standard, MES actuals as variance
Journals /finance/journals	Inventory valuation, payroll JV, depreciation run
Production Incentives /hrms/incentives	MES labour tracking and confirmed output
Labour Cost /hrms/labour-cost	MES labour hours priced by salary structure
Payroll Run /hrms/payroll	Attendance, overtime, leave, incentives, salary structure, statutory slabs

6. Step-by-step process flows

Eight flows. These are what you demonstrate at UAT, so they are also what you should be able to click through end to end at the close of every phase.

6.1 Procure to receipt

- 1 **Requirement appears.** MRP produces a buy suggestion, or a storekeeper raises a manual requisition from the Low Stock Monitor.
Planner / Storekeeper · /planning/mrp or /inventory/reorder
- 2 **Purchase requisition is created** with item, quantity, required-by date and a reason.
Buyer · /procurement/requisitions
- 3 **PR is approved** against the approval matrix. Below the value threshold this is auto-approve, and it still creates a workflow instance.
Purchase manager · /workflow/inbox
- 4 **RFQ is issued** to a supplier set drawn from the item's approved suppliers.
Buyer · /procurement/rfq
- 5 **Supplier quotations are captured**, one per supplier, with rate, tax, freight and lead time.
Buyer · /procurement/quotations
- 6 **Quotations are compared on landed cost**, not on rate. A recommendation is recorded with a reason.
Buyer · /procurement/comparison
- 7 **Purchase order is raised and approved.** The number comes from the numbering engine. After approval, changes go through amendment with a revision history, never in-place edit.
Buyer, then approver · /procurement/orders
- 8 **Material arrives; GRN is posted** against the PO, with batch or heat number captured for steel.
Storekeeper · /procurement/grn
- 9 **Incoming inspection.** Accepted quantity becomes usable stock; rejected quantity goes to a purchase return and a debit note.
QC inspector · /quality/inspections
- 10 **Put-away** moves accepted stock from the receiving bin to its storage bin. The stock ledger now shows it as available.
Storekeeper · /inventory/putaway

6.2 Plan to production order

- 1 **Forecast is entered** by month and SKU, with seasonality — demand is heavy in summer and the gifting season.
Planner · /planning/forecast
- 2 **Demand is consolidated** from the forecast plus firm orders (manual entry in release 1, sales orders in release 2).
Planner · /planning/demand
- 3 **Master schedule is firmed** for the planning horizon, bucketed by week.
Planner · /planning/mps
- 4 **MRP runs.** It explodes the BOM, nets against on-hand and reservations, treats open POs as scheduled receipts, applies lot-sizing and safety stock from the planning policy, and returns a buy list and a make list. This is a long-running job: it returns **202 Accepted** with a job uid.
Planner · /planning/mrp
- 5 **Capacity is checked** per work centre against the shift and holiday calendar. Overload is resolved by moving dates or adding a shift, before release.
Planner · /planning/capacity
- 6 **Production orders are created** from the make lines, and purchase requisitions from the buy lines.
Planner · /planning/orders
- 7 **Orders are sequenced** on the schedule board and material is reserved against them.
Planner · /planning/schedule, /planning/reservations
- 8 **Order is released to the shop floor.** Release is gated: no release without a valid BOM, a valid routing and reserved or available material.
Planner · /planning/orders

6.3 Order to output

- 1 Production order is received** on the shop floor and exploded into work orders, one per routing operation.
Production supervisor · /production/orders
- 2 Work orders queue at their work centre** in schedule sequence.
Supervisor · /production/queue
- 3 Material is issued** against the order from the raw material store.
Storekeeper · /inventory/issues
- 4 Operator confirms the operation:** good quantity, scrap quantity, rework quantity, machine, operator and time.
Consumption is backflushed from the BOM.
Operator · /production/entry
- 5 In-process inspection** at the operations that require it. A fail routes to rework or scrap, and scrap SS returns to stock as a recoverable item.
QC inspector · /quality/inspections
- 6 Repeat steps 4 and 5** down the routing: cutting, drawing, forming, necking, threading, welding, vacuum, leak test, polishing, coating, printing, assembly.
Operators · /production/entry
- 7 Downtime and losses are recorded** as they happen, which is what makes OEE real rather than decorative.
Operator · /production/downtime, /production/scrap
- 8 Final inspection**, including the temperature-retention test on a sample plan and a leak test at 100%.
QC inspector · /quality/inspections
- 9 Finished goods are received** into the FG store and the order is closed. The traveller card and genealogy now hold the full chain.
Supervisor · /production/wip, /production/genealogy

6.4 Pack to dispatch

- 1 Packing order is created** for the finished goods against a customer requirement.
Packing supervisor · /dispatch/packing-orders
- 2 Bottles are packed** into inner boxes, then cartons, then pallets. Each level gets a barcode label.
Packer · /dispatch/cartons, /dispatch/pallets, /dispatch/labels
- 3 Dispatch plan is built** for the day, grouping shipments by route and vehicle.
Dispatch officer · /dispatch/planning
- 4 Pick list is generated** and picked from the FG store.
Storekeeper · /dispatch/pick-lists
- 5 Vehicle is loaded** against the loading sheet, carton by carton.
Loader · /dispatch/loading
- 6 Shipment is created** with the delivery challan and, for domestic movement, the e-way bill.
Dispatch officer · /dispatch/shipments
- 7 Export orders** additionally raise export documents and a container-level packing list.
Export executive · /dispatch/exports, /dispatch/export-documents
- 8 Delivery is tracked and proof of delivery captured.** Freight cost closes against the shipment.
Dispatch officer · /dispatch/tracking, /dispatch/pod, /dispatch/freight

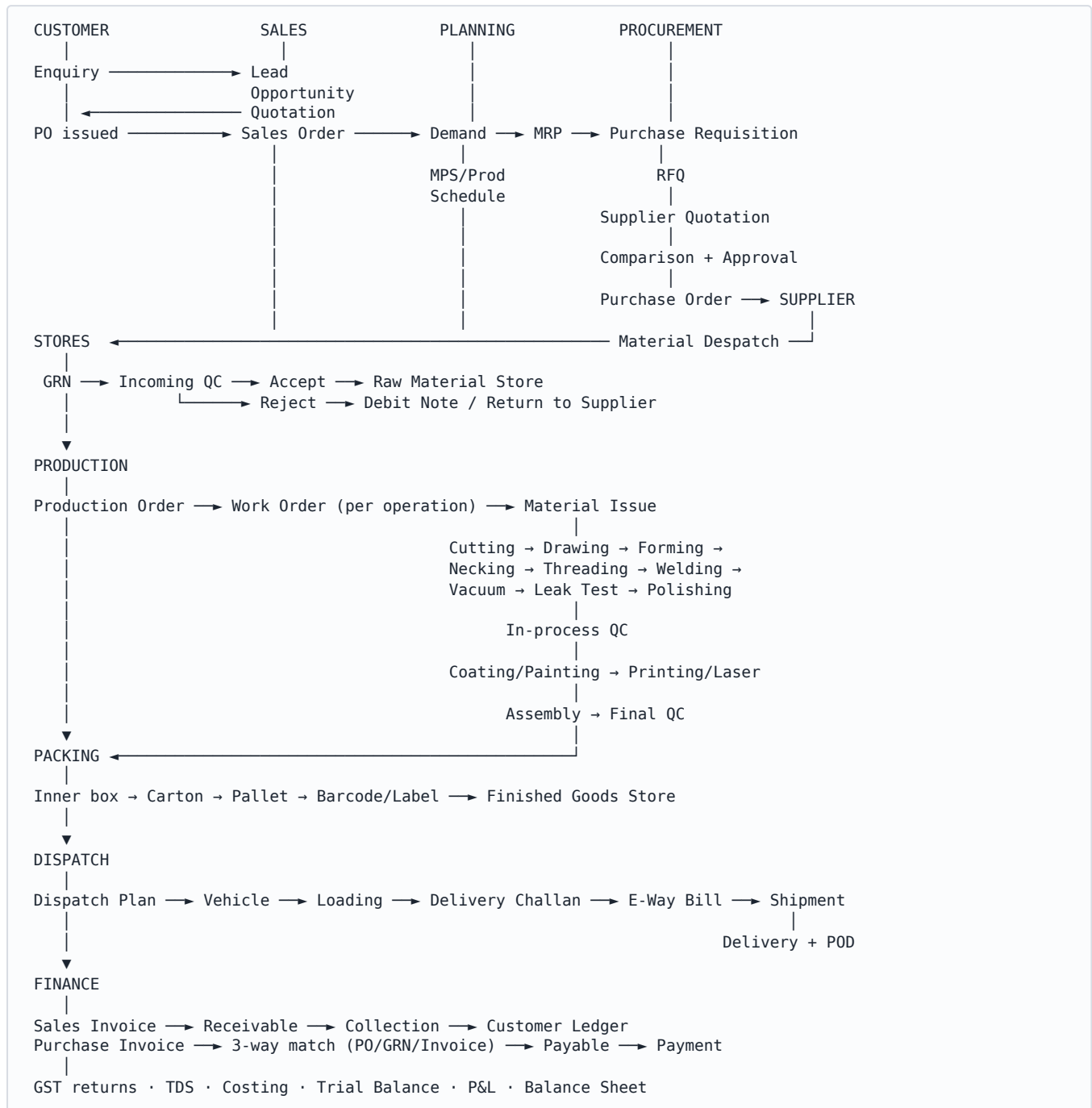
6.5 The other four flows, in brief

Flow	Steps
Record to report <i>Finance</i>	Purchase invoice arrives → three-way match against PO and GRN → payable posted → payment released → bank reconciliation. In parallel: shipment → sales invoice → receivable → collection → customer ledger. Then: inventory valuation posts to the GL, payroll posts a JV, depreciation runs, GST and TDS returns are prepared, period is closed, and the trial balance, P&L and balance sheet are produced.
Hire to pay <i>HRMS</i>	Manpower requisition → candidate pipeline → employee created → salary structure assigned → attendance captured daily → overtime and leave applied → incentives computed from confirmed output → payroll run → payslips published → statutory returns filed (PF, ESI, TDS) → payroll JV posted to Finance.

Flow	Steps
Maintain the plant <i>Maintenance</i>	Asset registered → preventive plan defined on calendar or meter → work order generated automatically or from a breakdown report → safety permit issued → spares issued from inventory → work completed and cost booked → condition readings trended → shutdown planned for the big jobs.
Set up the system <i>Admin</i>	Company and branches → plants, lines, warehouses and bins → departments and cost centres → financial year and periods → roles and permissions → users, with data scope → document numbering series → approval matrix and workflows → masters loaded by Excel import → opening stock and opening balances.

7. Overall project workflow

This is the whole business on one page, from the customer enquiry to the money in the bank. Everything in sections 4 to 6 is a slice of this diagram. It is taken from SRS Volume 0 §2.3, which is the approved baseline.



What the product is, physically

A vacuum-insulated bottle is an assembly, and the data model reflects it: an outer body and an inner body, each drawn from steel coil, joined by bottom welding, then vacuum-drawn and sealed, leak-tested, polished, coated, printed and assembled with a lid. One bottle model × capacity × colour × lid type × branding produces hundreds of SKUs from a few base bodies — which is why the product master is a variant model and not a flat SKU list, and why a body is generic stock until the coating operation commits it to one customer's branding.

8. Dependencies between portals

Dependencies decide the build order, and getting them wrong is the most expensive mistake available to us. A portal built before its upstream module exists gets built twice.

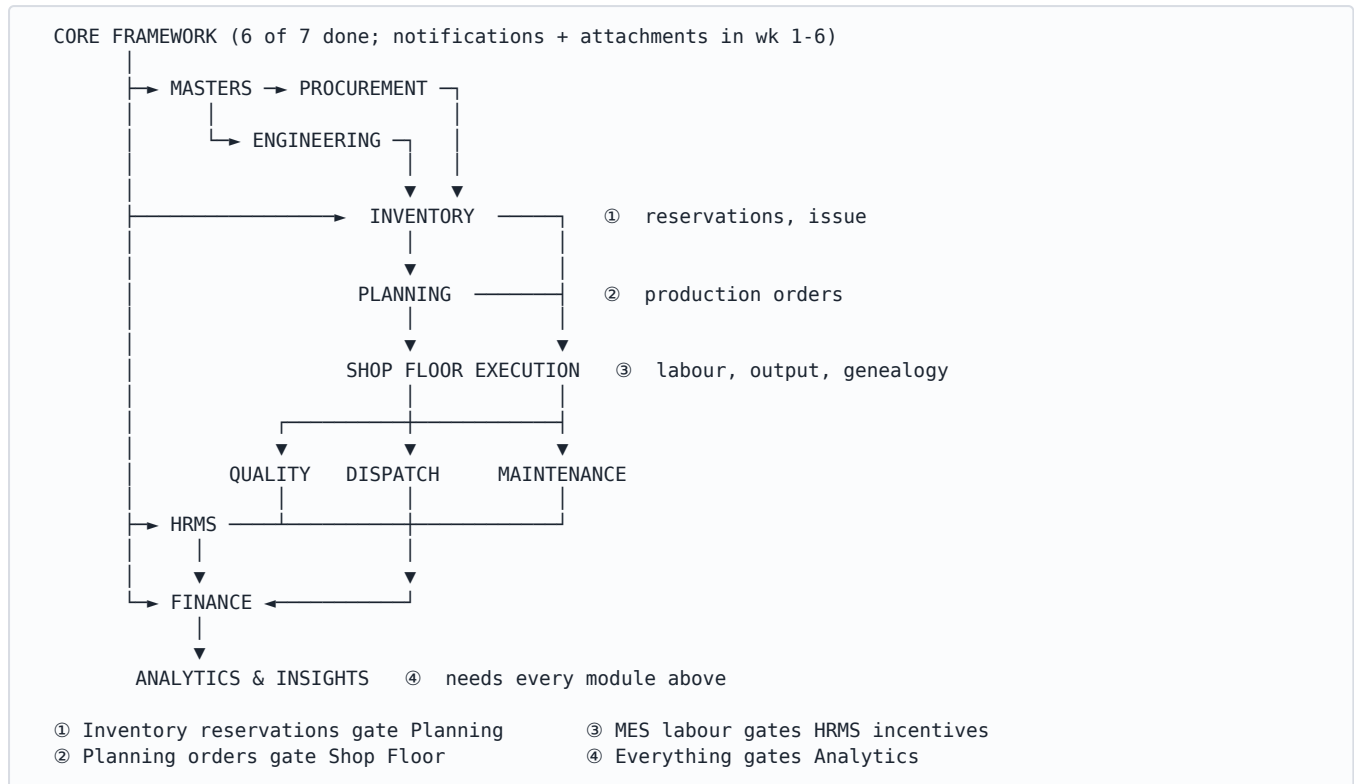
8.1 The dependency that applies to everything

Every document in every portal depends on all seven core framework services: authentication and RBAC, organisation and tenancy, document numbering, workflow and approvals, audit, notifications, and attachments. Six of the seven are built. **Notifications and attachments are the two gaps**, and they are the first thing the lead finishes, in weeks 1 to 6, because everything downstream needs them.

8.2 Portal dependency matrix

Portal	Depends on	Hard blocker?
Master Data	Core framework only	No — can start any time
Procurement	Masters (supplier, item, UOM, tax, payment terms), Inventory (GRN posts stock)	Both already built
Inventory & Stores	Masters (item, warehouse, bin), Procurement (GRN)	Both already built
Product Engineering	Masters (product items, work centres)	Already built
Production Planning	Engineering (BOM, routing, work centres), Masters (item, shift, holiday calendar, planning policy), Inventory (on-hand, reservations), Procurement (open POs as scheduled receipts)	Yes — needs Inventory reservations
Shop Floor Execution	Planning (production orders), Engineering (routing, operations), Inventory (material issue, FG receipt), Quality (in-process inspection), HRMS (shifts, labour)	Yes — needs Planning orders and Inventory issue
Quality	Procurement (GRN), Shop Floor (operations), Masters (parameters, defect types)	Built; final inspection needs MES
Plant Maintenance	Masters (machine), Inventory (spare parts issue), HRMS (technicians)	Soft — spares needs Inventory issue
Packing & Dispatch	Shop Floor (FG output), Inventory (FG stock), Masters (transporter), Sales (orders)	Built against stock, not MES
Finance & Accounts	Procurement (invoice, three-way match), Inventory (valuation to GL), Dispatch (sales invoice), HRMS (payroll JV), Masters (HSN, tax)	Soft — can build COA and journals first
HR & Payroll	Masters (employee), Shop Floor (labour hours, incentive output), Finance (payroll JV)	Soft — payroll before incentives
Analytics & Insights	Every module above, read-only	Yes — must be last
Sales & CRM	Masters (customer), Planning (demand), Dispatch, Finance	Blocked — no approved SRS

8.3 The four dependencies that actually shape the schedule



How to work across a dependency without waiting

Two rules from [CLAUDE.md](#) make parallel work possible. **Reading** across modules goes through the other module's published application service — never import another module's SQLAlchemy models, and never join across module tables in a write path. **Writing** across modules goes through domain events over the transactional outbox: an approved GRN does not call the inventory repository, it emits `procurement.grn.approved` and inventory subscribes. So agree the event name and payload on day one, and both sides can then build in parallel against it.

9. Development sequence

Six phases. They overlap on purpose — the manufacturing spine is serial, but Finance and HR are independent of it and run as a parallel track from week 3, which is what keeps the total at 18 weeks instead of 26.

Phase	What	Weeks	Why here and not elsewhere
0	Ramp-up and foundation Environment, codebase tour, standards, first small tickets	1–2	Five freshers cannot be productive on a 253-screen codebase on day one. Two weeks of deliberate ramp-up saves a month of rework.
1	Complete the operational spine Platform services, Master Data, Inventory, Production Planning	3–11	Inventory and Planning are on the critical path for Shop Floor. Platform services unblock everyone. Nothing downstream is safe to start until these are real.
2	Manufacturing execution Shop Floor Execution, then Plant Maintenance	6–18	Starts the moment Planning can release an order and Inventory can issue material. Maintenance follows because it consumes spares from Inventory and technicians from HR.
3	Commercial and people Finance & Accounts, HR & Payroll	3–16	Runs in parallel with phases 1 and 2 because neither depends on the manufacturing spine to <i>start</i> . Both are long, so they start early and finish late.
4	Analytics and hardening Analytics & Insights, plus hardening the complete portals	14–18	A dashboard over an unbuilt module shows nothing. Analytics can only be honest once its sources exist.
5	Integration, UAT, migration, go-live	15–18	Cross-portal flows, opening balances, opening stock, user training, cutover.

9.1 Recommended order of implementation, within any portal

This order is not optional. Every fresher who works back-to-front loses two days.

- 1 **Read the SRS volume, or write the functional baseline note if none exists.** Cite the requirement IDs. If the specification is silent, ask — do not invent a business rule.
- 2 **Design the data model.** Standard columns, soft delete with the `deleted_key` companion, correct decimal scales, indexes. Get it reviewed *before* writing the migration.
- 3 **Write the Alembic migration** and prove it runs up and down against a seeded database.
- 4 **Build the domain layer.** Entities, value objects, the state machine, business rules. No FastAPI import, no SQLAlchemy import, no I/O. Unit-test the rules here — this is where coverage is cheap.
- 5 **Build the application service:** use cases, orchestration, DTO mapping. One aggregate per transaction; if you must touch two, modify one and emit an event.
- 6 **Build the repository and models** in `infrastructure/`.
- 7 **Expose the API.** Pydantic v2 request and response schemas, router, and a declared permission dependency on every endpoint.
- 8 **Write the integration test** against real MySQL in Docker, never SQLite. Include the tenancy test and the 403-without-permission test.
- 9 **Now wire the front end.** Replace the mock import with the API client, keep the layout, delete the mock fixture in the same commit.
- 10 **Walk the flow end to end** in the browser, from the screen that precedes yours to the screen that follows it.

10. Team allocation

Six people: the lead plus five developers. Fill in the Name column at the first standup so this table is the single answer to "who owns that screen".

Code	Name	Primary portal	Then	Why this pairing
Lead	_____	Administration & Platform Services <i>13 screens</i>	Integration, release, UAT <i>wk 13–18</i>	Platform services unblock every other portal, and they are where architectural decisions concentrate. Half your time; the other half is review, coordination and unblocking.
D1	_____	Inventory & Stores <i>21 screens</i>	Plant Maintenance & EAM <i>12 screens, wk 12–18</i>	Maintenance spares are issued from Inventory. Whoever built the issue transaction should build the screen that consumes it.
D2	_____	Master Data (4) then Production Planning (8)	Shop Floor Execution <i>with D3, wk 12–14</i>	Masters is the gentlest start in the codebase — a good first three weeks. Planning then follows naturally, and its production orders are what Shop Floor consumes, so D2 moves to MES already knowing the contract.
D3	_____	Shop Floor Execution <i>17 screens, lead developer</i>	Analytics & Insights <i>with D4, wk 15–18</i>	The largest single piece of work, so it gets the longest run. Weeks 3–5 are spent hardening Procurement and Engineering, which is the fastest way to learn the patterns MES must follow.
D4	_____	Finance & Accounts <i>12 screens</i>	Analytics: Finance and Sales <i>wk 14–18</i>	Finance needs one owner for the whole double-entry model — splitting it across two people produces two chart-of-accounts designs. Extra review time from the lead is budgeted here.
D5	_____	HR, Payroll & Workforce <i>22 screens</i>	UAT and training material <i>wk 17–18</i>	Biggest portal by screen count, and largely self-contained, so one person can hold the whole picture. D4 joins from week 14 for the payroll journal voucher, which is where HR meets Finance.

10.1 Screen-level allocation for the two shared portals

Portal	Owner	Screens
Shop Floor Execution	D3	Production orders, Work orders, Work centre queue, Production entry , WIP, Machine status, Downtime, OEE , Traveller card, Genealogy, Dashboard
	D2	Work instructions, Scrap, Rework, Shift management, Labour tracking, Production reports
Analytics & Insights	D3	Executive overview, Production, Quality, Maintenance, Inventory analytics, Alerts & exceptions, KPI library, Data sources
	D4	Dashboards, AI insights, Procurement, Sales, Finance, Workforce analytics, Forecasts, Scorecards, Report catalogue, Access & audit

One module per branch

Branch per module, not per day and not per three modules. Two people must never be editing the same module's [domain/](#) in the same week — that is what the screen-level split above exists to prevent. Cite requirement IDs in the commit message.

11. Responsibilities of each team member

11.1 Lead you

Area	Responsibility
Build	Administration & Platform Services: notifications, attachments and DMS, document repository, import and export framework, collaboration, barcode and labels, report engine, master data framework, monitoring, jobs, backup, licence, integrations. 13 screens, half-time, weeks 1 to 12.
Architecture	Own the data model review for every portal before its migration is written. Own the ADR register. Say no to a new dependency. Decide every cross-module event name and payload.
Review	Review every pull request. Nothing merges without your approval. Budget 1.5 hours a day for this and protect it — with five freshers, review is not overhead, it is how the codebase stays coherent.
Specification	Write the functional baseline note for each portal that has no SRS volume (section 15.1). Keep open-questions.md current. Take every "the spec is silent" question to the client and come back with an answer, not a guess.
Coordination	Daily 15-minute standup. Weekly Friday demo of working software. Weekly re-forecast against section 13. Unblock before you build — a blocked developer costs more than your half-day of code.
Release	CI, migrations, environments, cutover plan, opening balances, opening stock, training material, go-live.

11.2 Every developer, D1 to D5

Area	Responsibility
Own your portal end to end	Data model, migration, domain, application service, repository, API, tests, and the screen. You are not a front-end developer or a back-end developer on this team. You own vertical slices.
Follow the ten-point checklist	Section 14. A transaction document missing any of the ten is not done, however good the screen looks.
Test what you build	Unit test for every business rule and state transition. Integration test for every endpoint against real MySQL. The tenancy test and the 403 test are not optional.
Never invent a business rule	If the SRS is silent, raise it with the lead the same day and record it in open-questions.md . Manufacturing and statutory logic guessed wrong is worse than absent.
Delete the mock	When your screen goes live against the API, delete its fixture in the same commit. A mock left behind becomes a second source of truth.
Demo on Friday	Show working software in the browser, not a pull request. If you cannot demo it, it is not done.
Ask on day one, not day three	You are all new to this. The expensive failure mode is a fresher quietly guessing for three days. A question costs ten minutes.

11.3 Individual accountabilities

Who	Accountable for, specifically
D1	The stock ledger stays the single truth for quantity and value: every movement writes exactly one ledger row, and valuation reconciles to the general ledger. Maintenance work orders generate correctly from both calendar and meter triggers.
D2	The generic master registry stays the way simple masters are added. MRP output reconciles to demand: no phantom shortages, no double-counted scheduled receipts. The production-order release gate never releases without BOM, routing and material.
D3	Production entry is one atomic transaction: output, backflushed consumption, scrap and rework either all post or none do. OEE matches a hand calculation for one machine over one shift. Genealogy resolves heat number to bottle in both directions.
D4	Debits equal credits, always. The three-way match cannot be bypassed. GST and TDS rates live in master data behind the statutory/ adapter, never in code. The trial balance ties to the sum of ledger rows.
D5	The payroll run is reproducible and reversible, and every payslip figure is traceable to attendance, structure and slab. Statutory returns match the filed formats. No salary or bank detail ever reaches a log or an audit row.

12. Duration estimates

Person-days of remaining effort, by portal, using the type rates from section 1.3. Calendar weeks assume the owner works on it as their primary task, with the usual meeting and review overhead already absorbed.

Portal / workstream	Screens left	Person-days	Owner	Weeks	Dominant cost
Ramp-up and foundation	—	—	All	1–2	Learning, not building
Administration & Platform	13	30	Lead	1–12	Import framework, report engine, notifications
Master Data	4	10	D2	3–5	Duplicate finder, Excel import
Inventory & Stores	21	40	D1	3–11	Issue, transfer, traceability, job work
Production Planning	8	25	D2	6–11	MPS, schedule board, forecast seasonality
Shop Floor Execution	17	50	D3 + D2	6–14	Production entry, OEE, genealogy
Plant Maintenance & EAM	12	32	D1	12–18	Preventive-plan scheduler
Finance & Accounts	12	45	D4	3–13	GST and TDS, statements, three-way match
HR, Payroll & Workforce	22	58	D5 + D4	3–16	Payroll engine, attendance import, statutory
Analytics & Insights	18	35	D3 + D4	14–18	Cross-module SQL in the reporting module
Hardening: Procurement, Engineering, Quality, Packing & Dispatch	—	18	Rotating	3–5, 18	Missing tests, edge cases, print templates
Integration, UAT, data migration, go-live	—	40	All	15–18	Cross-portal flows, opening balances, training
Release 1 total	127	383	6 people	18	

12.1 Does the arithmetic close?

Check	Person-days	Working
Work required	383	Sum of the table above
Lead capacity	45	18 weeks at 50% of a 5-day week
Developer capacity	400	5 developers × 16 build weeks (3–18) × 5 days
Total capacity	445	16% slack over the requirement

16% slack is the right amount for a mostly-fresher team. It absorbs the two things that always happen: rework after review, and one portal turning out harder than it looked. It does not absorb scope growth, which is why section 15 matters.

12.2 The confidence band

Scenario	Go-live	What has to be true
Optimistic	16 weeks	The team ramps faster than budgeted, no specification gaps block Finance or HR, and the mock fixtures turn out to be accurate contracts.
Plan	18 weeks	The schedule in section 13, as written. This is the number to commit to.
Realistic worst case	21 weeks	Payroll or GST needs a rebuild after review, or two developers lose a week each to the same misunderstood pattern.

Release 2 — Sales & CRM, the Shop Floor Terminal, the Supplier and Customer portals, and the mobile app — is a further **8 to 10 weeks** after go-live, and cannot be scheduled until SRS Volume 2 is written and approved.

13. The 18-week schedule

One row per workstream, one column per week. Bars are coloured by phase; the owner and the week range are printed on every row, and section 12 carries the same data as a table.

Workstream	Owner	Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Ramp-up & foundation	All	1–2																		
Administration & Platform	Lead	1–12																		
Master Data completion	D2	3–5																		
Inventory & Stores	D1	3–11																		
Production Planning	D2	6–11																		
Shop Floor Execution	D3+D2	6–14																		
Plant Maintenance & EAM	D1	12–18																		
Finance & Accounts	D4	3–13																		
HR, Payroll & Workforce	D5+D4	3–16																		
Analytics & Insights	D3+D4	14–18																		
Hardening, 4 complete portals	Rotate	3–5, 18																		
Integration, UAT, go-live	All	15–18																		

■ Phase 0 · Ramp-up
 ■ Phase 1 · Operational spine
 ■ Phase 2 · Manufacturing execution
 ■ Phase 3 · Commercial & people
 ■ Phase 4/5 · Analytics, hardening, go-live

13.1 Milestones

Each milestone is a demonstrable state of the product, not a percentage. If it cannot be clicked through in the browser at the Friday demo, it has not been met.

End of week	Milestone	Means
2	Team ready	All six environments run the stack. Every developer has merged one small screen. CI is green and the review loop works.
5	Masters closed	Master Data is 100% complete. Procurement and Product Engineering are hardened. Notifications and attachments are live for every portal to use.
8	Stock moves	Inventory issue, return and transfer post real ledger rows. Finance chart of accounts and journals post double-entry.
11	Plan to order	Inventory and Production Planning complete. Flow 6.2 demonstrable end to end: forecast to MRP to a released production order.
13	Platform closed	Administration & Platform Services complete. Finance complete except period close. Production entry live on the shop floor.
14	Factory runs	Shop Floor Execution complete. Flow 6.3 demonstrable end to end: released order to finished goods, through quality gates.
16	People paid	HR & Payroll complete, with one full payroll run reproduced and reversed. Analytics well advanced.
17	Feature complete	Plant Maintenance and Analytics complete. All eight flows in section 6 demonstrable.
18	Go-live	UAT signed off. Opening balances and opening stock loaded. Users trained. Cutover.

13.2 The weekly rhythm

When	What
Daily, 15 minutes	Standup. Yesterday, today, blocked on what. A blocker named at standup is the lead's problem that day, not the developer's problem all week.
Continuously	Pull requests, reviewed by the lead. Nothing merges unreviewed. One module per branch.
Friday, 1 hour	Demo of working software in the browser, each developer in turn. Then the lead re-forecasts the schedule and says out loud whether week 18 still holds.

When	What
Fortnightly	Walk one complete flow from section 6 end to end, across portals, with the owners of every portal it touches in the room.

14. Definition of done

The point of this section is that "the screen looks right" is not done. These lists are taken from the standing rules in `CLAUDE.md`, and they are what the lead reviews against.

14.1 Every transaction document — all ten, no exceptions

A transaction document is anything with a document number and a lifecycle: a purchase order, a material issue, a production order, an inspection, a work order, a payroll run. If you are adding one and any of these ten is missing, the work is not done.

#	Requirement	What it means concretely
1	Attachments	Through <code>core_attachment</code> , polymorphic on <code>entity_type</code> plus <code>entity_id</code> .
2	Comments thread	Through <code>core_comment</code> , same polymorphic key, @mention capable.
3	Approval workflow	A <code>core_workflow_instance</code> exists even when the configured workflow is auto-approve.
4	Configurable document number	From the numbering engine. Never a raw sequence in module code.
5	Standard lifecycle	DRAFT, PENDING_APPROVAL, APPROVED, IN_PROGRESS, COMPLETED, with REJECTED, ON_HOLD, CANCELLED, CLOSED, AMENDED as applicable. Validated by an explicit state machine, never by scattered <code>if</code> checks.
6	Amendment and revision	Full version history and a diff view. Never an in-place edit after approval.
7	Cancellation with a reason code	Mandatory reason. Never deletion.
8	Print and PDF	With a configurable template.
9	Audit log entries	On create, update, submit, approve, reject, cancel, amend, delete.
10	Domain events	Emitted on every state change, through the transactional outbox.

14.2 Every master

Configurable without a code change, and carrying: a unique `code` (optionally auto-numbered), a `name`, `is_active`, effective-dated activation where relevant, attachments, notes, an audit trail, Excel import, export, and a **where-used dependency check before deactivation**. A master is never hard-deleted, and may not be deactivated while an open transaction references it.

14.3 Quality gates, enforced in CI

Gate	Requirement
Unit tests	Every domain rule, calculation and state transition. <code>pytest</code> .
Integration tests	Every API endpoint, against real MySQL in Docker. Never SQLite.
Coverage	≥85% on <code>domain/</code> and <code>application/</code> ; ≥70% overall.
Contract tests	The OpenAPI schema is generated, committed, and diffed in CI.
Migration tests	Every Alembic revision runs up and down against a seeded database.
Tenancy tests	Each module proves that a cross-company read returns nothing.
RBAC tests	Each endpoint returns 403 without its permission.
Front end	Vitest and React Testing Library units; Playwright for the critical journeys.
Lint and types	<code>ruff</code> and <code>mypy --strict</code> on the backend; ESLint, Prettier and <code>tsc</code> on the front end.

Calculations that may never ship on "looks right"

Financial posting, inventory valuation, BOM explosion, MRP and payroll all require test cases derived from worked examples. If there is no worked example, there is no test, and if there is no test the calculation does not ship. This is the single rule most likely to save this project from an expensive go-live.

15. Risks and open items

Eight risks, ordered by what they cost us. Each has a named mitigation, and the mitigation is work that is already inside the plan — not a hope.

R1. Seven of eleven SRS domain volumes are unwritten High impact · Certain

Volumes 5 to 11 are "Not started", yet Planning, Shop Floor, Quality, Dispatch, Finance and HRMS are all being built. [CLAUDE.md](#) forbids coding before the volume exists, so today we are formally out of process on most of the remaining work.

Mitigation. The lead writes a **3 to 5 page functional baseline note** per remaining portal before its build starts — scope, document lifecycle, business rules, worked examples — rather than a full 14-chapter volume. Half a day each, nine notes, budgeted inside the lead's weeks 1 to 12. Anything still unresolved goes to [open-questions.md](#) as a numbered assumption.

R2. Payroll and GST have no worked examples High impact · Likely

Statutory calculation guessed wrong is not a bug, it is a legal exposure. Neither the payroll engine nor the GST and TDS logic has a single worked example in the repository to test against.

Mitigation. Before week 8, get three real payslips and one filed GST return from the client's accountant. Turn them into fixed test cases. D4 and D5 do not write the engine until those cases exist. Rates and slabs live in master data behind the [statutory/](#) adapter.

R3. Fresher velocity is the largest unknown High impact · Likely

Five developers new to a 253-screen codebase. A single misunderstood pattern repeated across twenty screens costs a fortnight.

Mitigation. 16% schedule slack. A weekly Friday demo of working software, not of pull requests. Re-forecast every Friday against section 13 and move a portal owner before a slip becomes three weeks. The lead reviews every pull request, and reviews the data model *before* the migration.

R4. Sales & CRM is absent but two portals need it Medium impact · Certain

Planning normally nets against sales orders; Finance receivables normally starts from a sales invoice. Neither exists in release 1.

Mitigation. Planning takes demand from the forecast plus manual firm-order entry. Finance raises the sales invoice from the dispatch shipment. Both are documented as temporary in [open-questions.md](#) and both are replaced in release 2.

R5. The mock fixtures are a hint, not a contract Medium impact · Possible

Eighty screens have mock data shaped for the convenience of the UI. A payload designed to make a table render is often not a sane API response.

Mitigation. Before designing a model, read the mock *and* the relevant SRS section, then design the API from the domain. Where they disagree, the domain wins and the screen changes. Never let a mock shape a database table.

R6. Cross-module coupling creeps in Medium impact · Likely

The fastest way for a fresher to finish a screen is to import another module's SQLAlchemy model and join across it. That one shortcut dissolves the modular monolith.

Mitigation. import-linter in CI enforces the dependency direction. Read across modules only through the other module's application service; write across modules only through domain events on the transactional outbox. Agree the event name and payload on day one.

R7. Finance has a single owner Medium impact · Possible

D4 alone holds the double-entry model, the three-way match and the statutory adapter. If D4 stalls or leaves, 45 person-days of the most correctness-critical work has no second reader.

Mitigation. D4 pairs with the lead for one session a week through weeks 3 to 13. The lead personally reviews the chart of accounts and the posting rules. D5 shadows the payroll journal voucher from week 14, which gives a second person context.

R8. UAT generates new scope Medium impact · Certain

Users see working software in week 15 and immediately want changes. Absorbing them silently is how an 18-week plan becomes 30 weeks.

Mitigation. Every UAT finding is triaged into go-live blocker or release 2. Only a blocker enters the schedule, and only by displacing something else. The lead owns that call, in writing.

15.1 What the lead should do in week 1

- 1 **Fill in the names** in the section 10 table and circulate this handbook.
- 2 **Start the nine functional baseline notes** (risk R1), in build order: Inventory first, because D1 starts on it in week 3.
- 3 **Ask the client for the payroll and GST worked examples** (risk R2). This has the longest lead time of anything in the plan, so it goes out in week 1, not week 7.
- 4 **Turn on import-linter in CI** (risk R6) before five freshers start writing modules.
- 5 **Record the two Sales & CRM substitutions** in [open-questions.md](#) (risk R4) so nobody re-litigates them in week 12.
- 6 **Agree the cross-module event names** for GRN posting, production confirmation, FG receipt, valuation posting and payroll journal voucher. Five names, written down, and Finance, Inventory, MES and HR can then build in parallel.

Appendix A — Full screen and route reference

All 253 routes in the product, grouped by portal, generated from `frontend/src/App.tsx` and the generic master registry. The label column is the sidebar caption from `navigation.ts`; a blank label means the screen is reachable by URL or from another screen rather than from the sidebar. Use this as the checklist when you estimate or hand over a portal.

Administration & Platform Services — 35 screens Lead

Route	Sidebar label	Route	Sidebar label
<code>/admin/admin-reports</code>	Administration reports	<code>/admin/license</code>	Licence
<code>/admin/api-keys</code>	API keys	<code>/admin/login-activity</code>	Login activity
<code>/admin/attachments</code>	Attachments & DMS	<code>/admin/masters</code>	Master data framework
<code>/admin/audit</code>	Audit trail	<code>/admin/monitoring</code>	Logging & monitoring
<code>/admin/backup</code>	Backup & restore	<code>/admin/notifications</code>	Notifications
<code>/admin/barcode</code>	Barcode & labels	<code>/admin/numbering</code>	Document numbering
<code>/admin/branches</code>	Branches	<code>/admin/organisation</code>	Structure explorer
<code>/admin/collaboration</code>	Collaboration	<code>/admin/parameters</code>	System parameters
<code>/admin/company</code>	Company	<code>/admin/permissions</code>	Permission explorer
<code>/admin/cost-centres</code>	Cost centres	<code>/admin/plants</code>	Plants
<code>/admin/currency</code>	Currency & rates	<code>/admin/reports</code>	Report engine
<code>/admin/delegations</code>	Delegations	<code>/admin/roles</code>	Roles
<code>/admin/departments</code>	Departments	<code>/admin/security</code>	Security policy
<code>/admin/documents</code>	Document repository	<code>/admin/sessions</code>	Sessions & devices
<code>/admin/financial-year</code>	Financial year & periods	<code>/admin/sod</code>	Segregation of duties
<code>/admin/import</code>	Import & export	<code>/admin/users</code>	Users
<code>/admin/integrations</code>	Integrations	<code>/admin/warehouses</code>	Warehouses & bins
<code>/admin/jobs</code>	Background jobs		

Administration — Workflow — 4 screens Lead

Route	Sidebar label	Route	Sidebar label
<code>/workflow/designer</code>	Workflow designer	<code>/workflow/matrix</code>	Approval matrix
<code>/workflow/inbox</code>	Approvals inbox	<code>/workflow/monitor</code>	Workflow monitor

Master Data — 38 screens D2

Route	Sidebar label	Route	Sidebar label
<code>/masters</code>	Dashboard	<code>/masters/machine</code>	Machines
<code>/masters/bank</code>	Banks	<code>/masters/packaging</code>	Packaging materials
<code>/masters/bottle-capacity</code>	Capacities	<code>/masters/payment-terms</code>	Payment terms
<code>/masters/bottle-colour</code>	Colours & finishes	<code>/masters/planning-policy</code>	Planning policy
<code>/masters/bottle-model</code>	Bottle models	<code>/masters/product</code>	Products
<code>/masters/city</code>	Cities	<code>/masters/product-category</code>	Product Category
<code>/masters/company-items</code>	Company Items	<code>/masters/product-items</code>	Product Items
<code>/masters/contact</code>	Contact persons	<code>/masters/production-lines</code>	Production lines

Route	Sidebar label	Route	Sidebar label
/masters/country	Countries	/masters/quality-parameter	Inspection parameters
/masters/customer	Customers	/masters/reason-code	Reason codes
/masters/defect	Defect types	/masters/shift	Shifts
/masters/duplicates	—	/masters/state	States
/masters/employee	Employees	/masters/steel-grade	Steel grades
/masters/holiday-calendar	Holiday calendar	/masters/steel-thickness	Steel thickness
/masters/hsn	HSN / SAC codes	/masters/supplier	Suppliers
/masters/import	—	/masters/tax	Tax rates
/masters/item	—	/masters/transporter	Transporters
/masters/item-category	Item Category	/masters/uom	Units of measure
/masters/lid-type	Lid & cap types	/masters/work-centres	Work centres

Procurement — 8 screens complete

Route	Sidebar label	Route	Sidebar label
/procurement	Dashboard	/procurement/orders	Purchase Order
/procurement/approvals	Approvals	/procurement/quotations	Vendor Quotations
/procurement/comparison	Quotation Comparison	/procurement/requisitions	Purchase Requisition
/procurement/grn	Goods Receipt Note	/procurement/rfq	Request for Quotation

Inventory & Stores — 31 screens D1

Route	Sidebar label	Route	Sidebar label
/inventory	Dashboard	/inventory/reorder	Low Stock Monitor
/inventory/abc-xyz	—	/inventory/reports	—
/inventory/adjustments	—	/inventory/requisitions	—
/inventory/ageing	—	/inventory/reservations	—
/inventory/batches	Batch & Expiry	/inventory/returns	Stock Return
/inventory/bin-map	—	/inventory/scrap	—
/inventory/category-ledger	Category Ledger	/inventory/serials	—
/inventory/counting	—	/inventory/stock	Current Stock
/inventory/goods-in-transit	—	/inventory/structure	—
/inventory/issues	Stock Out	/inventory/traceability	—
/inventory/job-work	—	/inventory/transfers	Stock Transfer
/inventory/labels	—	/inventory/valuation	—
/inventory/ledger	Stock Ledger	/inventory/variance	—
/inventory/movement	—	/inventory/verification	—
/inventory/putaway	—	/inventory/warehouses	Warehouse Setup
/inventory/receipts	Stock In		

Product Engineering — 13 screens complete

Route	Sidebar label	Route	Sidebar label
/engineering	Dashboard	/engineering/operations	Operations

Route	Sidebar label	Route	Sidebar label
/engineering/bom	BOM register	/engineering/products	—
/engineering/bom-explorer	BOM explorer	/engineering/reports	Engineering reports
/engineering/changes	Engineering changes	/engineering/routing	Manufacturing routing
/engineering/costing	Cost roll-up	/engineering/tools	Tools & dies
/engineering/document-types	Document types	/engineering/work-centres	Work centres
/engineering/documents	Engineering documents		

Production Planning — 11 screens D2

Route	Sidebar label	Route	Sidebar label
/planning	Dashboard	/planning/mrp	MRP
/planning/calendar	Production Calendar	/planning/orders	Production Orders
/planning/capacity	Capacity Planning	/planning/reports	Reports
/planning/demand	Demand	/planning/reservations	Material Reservation
/planning/forecast	Sales Forecast	/planning/schedule	Schedule Board
/planning/mps	Master Schedule		

Shop Floor Execution — 17 screens D3 + D2

Route	Sidebar label	Route	Sidebar label
/production	Dashboard	/production/queue	Work centre queue
/production/downtime	Downtime	/production/reports	Production reports
/production/entry	Production entry	/production/rework	Rework
/production/genealogy	Production genealogy	/production/scrap	Scrap
/production/instructions	Work instructions	/production/shifts	Shift management
/production/labour	Labour tracking	/production/traveller	Traveller card
/production/machines	Machine status	/production/wip	Work in progress
/production/oeo	OEE	/production/work-orders	Work orders
/production/orders	Production orders		

Quality — 11 screens complete

Route	Sidebar label	Route	Sidebar label
/quality	Dashboard	/quality/inspections	Inspections
/quality/audits	Audits	/quality/ncr	Non-Conformance
/quality/calibration	Calibration	/quality/plans	Inspection Plans
/quality/capa	CAPA	/quality/reports	Reports
/quality/complaints	Customer Complaints	/quality/suppliers	Supplier Quality
/quality/defects	Defects & Pareto		

Plant Maintenance & EAM — 12 screens D1

Route	Sidebar label	Route	Sidebar label
/maintenance	Dashboard	/maintenance/reports	Reports
/maintenance/assets	Asset Register	/maintenance/shutdowns	Shutdowns

Route	Sidebar label	Route	Sidebar label
/maintenance/breakdowns	Breakdowns	/maintenance/spares	Spare Parts
/maintenance/condition	Condition Monitoring	/maintenance/technicians	Technicians
/maintenance/permits	Safety Permits	/maintenance/utilities	Utilities
/maintenance/pm	Preventive Plans	/maintenance/work-orders	Work Orders

Packing, Dispatch & Logistics — 19 screens complete

Route	Sidebar label	Route	Sidebar label
/dispatch	Dashboard	/dispatch/pick-lists	Pick lists
/dispatch/cartons	Cartons	/dispatch/planning	Dispatch planning
/dispatch/export-documents	Export documents	/dispatch/pod	Proof of delivery
/dispatch/exports	Export shipments	/dispatch/reports	Dispatch reports
/dispatch/freight	Freight & cost	/dispatch/returns	Returns
/dispatch/labels	Labels	/dispatch/shipments	Shipments
/dispatch/loading	Vehicle loading	/dispatch/tracking	Shipment tracking
/dispatch/packing-materials	Packaging material	/dispatch/transporters	Transporter performance
/dispatch/packing-orders	Packing orders	/dispatch/vehicles	Vehicles
/dispatch/pallets	Pallets		

Finance & Accounts — 12 screens D4

Route	Sidebar label	Route	Sidebar label
/finance	Dashboard	/finance/costing	Cost & Variance
/finance/accounts	Chart of Accounts	/finance/journals	Journals & Ledger
/finance/assets	Fixed Assets	/finance/payables	Payables
/finance/banking	Banking	/finance/receivables	Receivables
/finance/budgets	Budgets	/finance/statements	Statements
/finance/close	Period Close	/finance/tax	GST & TDS

HR, Payroll & Workforce — 22 screens D5 + D4

Route	Sidebar label	Route	Sidebar label
/hrms	Dashboard	/hrms/performance	Performance
/hrms/attendance	Attendance	/hrms/recruitment	Candidate pipeline
/hrms/contractors	Contractor labour	/hrms/reports	HR reports
/hrms/employees	Employees	/hrms/requisitions	Manpower requisitions
/hrms/incentives	Production incentives	/hrms/salary-structures	Salary structures
/hrms/labour-cost	Labour cost	/hrms/self-service	My page
/hrms/leave	Leave	/hrms/shifts	Shifts & roster
/hrms/organisation	Organisation structure	/hrms/skills	Skill matrix
/hrms/overtime	Overtime	/hrms/statutory	Statutory compliance
/hrms/payroll	Payroll run	/hrms/team	My team
/hrms/payslips	Payslips	/hrms/training	Training

Analytics & Insights — 18 screens D3 + D4

Route	Sidebar label	Route	Sidebar label
/bi	Executive Overview	/bi/kpis	KPI library
/bi/alerts	Alerts & Exceptions	/bi/maintenance	Maintenance
/bi/dashboards	Dashboards	/bi/procurement	Procurement
/bi/data-sources	Data sources	/bi/production	Production
/bi/finance	Finance	/bi/quality	Quality
/bi/forecasts	Forecasts	/bi/reports	Report Catalogue
/bi/governance	Access & audit	/bi/sales	Sales & customers
/bi/insights	AI Insights	/bi/scorecards	Scorecards
/bi/inventory	Inventory	/bi/workforce	Workforce

Sign-in — 1 screens complete

Route	Sidebar label	Route	Sidebar label
/login	—		

User profile — 1 screens complete

Route	Sidebar label	Route	Sidebar label
/profile	—		

Keep this handbook current

Sections 3, 12 and 13 go stale the moment a portal lands. Re-measure and re-issue at the end of every phase — weeks 2, 11, 14 and 18. The status in section 3 was produced by counting routes in `App.tsx` and counting page components that import from `@/api`, so it takes minutes to redo, and a handbook nobody trusts is worse than none.